

MDS-UPDRS Finger Tapping Analysis

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a subject performing the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb 10 times as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 consecutive finger taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested.

Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide your score on the last line.

EXTRACTED FEATURES

Number of taps: 11

Tap periods (frames): [187, 300, 207, 185, 200, 146, 154, 138, 110, 87]

Tap amplitudes (pixels): [110.43550153822818, 100.70253224224304, 117.33712115098103, 112.36102527122117, 109.9317970379817, 101.24228365658294, 83.6301381082203, 104.04326023342406, 103.60019305001319, 109.01834707974616, 102.07840124139877]

GPT-5

MDS-UPDRS 3.4 Finger Tapping — Scoring Analysis

Given data

- Number of taps detected: 11 (10 instructed; 11 suggests an extra detected tap, but sequence length allows analysis of rhythm, speed, and amplitude trends)
- Tap periods (frames): [187, 300, 207, 185, 200, 146, 154, 138, 110, 87] → 10 intervals (between 11 peaks)
- Tap amplitudes (pixels): [110.44, 100.70, 117.34, 112.36, 109.93, 101.24, 83.63, 104.04, 103.60, 109.02, 102.08] → 11 taps

Note: Without an explicit frame rate, we assess relative speed (slowing) and rhythm from the sequence of intervals and interruptions from outliers/large pauses.

Rhythm and interruptions

- Periods show one clearly prolonged interval early: 300 (between taps 1–2 or 2–3 depending on indexing), followed by more regular and faster intervals.
- Large-to-small trend in periods: 187 → 300 → 207 → 185 → 200 → 146 → 154 → 138 → 110 → 87 (overall speeding up after the early long interval).
- Interruptions/hesitations: The lone 300-frame interval likely represents a single hesitation. There are no repeated long pauses and no evidence of >3 distinct interruptions. No clear long arrest (freeze).

Interpretation: 1–2 interruptions at most.

Speed

- After the early outlier, periods generally decrease (i.e., tapping speeds up), which does not indicate slowing. If anything, speed improves through the sequence. There is no pattern of mild or moderate slowing across the sequence.

Interpretation: No consistent slowing.

Amplitude and decrement

- Amplitudes across 11 taps: 110.4, 100.7, 117.3, 112.4, 109.9, 101.2, 83.6, 104.0, 103.6, 109.0, 102.1
- There is variability, but not a monotonic decrement from the start:
 - Early taps are large; a dip at tap 7 (83.6) is followed by recovery (104–109) and ends around ~102.
 - This pattern is not “decrement starting after the 1st tap” (score 3 criterion), nor is it a clear “decrement midway” that persists; it’s a transient dip with recovery.
 - A slight end-of-sequence lowering vs. the largest values could be interpreted as slight decrement near the end, but the presence of recovery and overall variability argues against a sustained decrement pattern.

Interpretation: At most a slight decrement near the end; not a sustained mid-sequence decrement.

Mapping to criteria

- 0 (Normal): Not applicable—there is at least one notable hesitation (the 300-frame interval).
- 1 (Slight): Fits well due to:
 - “Regular rhythm broken with one or two interruptions or hesitations” → matches the single prominent hesitation.
 - “Slight slowing” → not present, but only one of the slight criteria is needed.
 - “Amplitude decrements near the end” → arguably minimal; variability with mild reduction at the end could be considered slight at most.
- 2 (Mild): Would require 3–5 interruptions, mild slowing, or amplitude decrement midway. Not supported by the data (no 3–5 hesitations; no slowing; no sustained mid-sequence decrement).
- 3 (Moderate): Not supported (no >5 interruptions, no long arrest, no moderate slowing, no decrement starting after 1st tap).
- 4 (Severe): Clearly not the case.

Final score: 1
